episodes. Imagine how many episodes an autonomous vehicle would need to sufficiently explore its state space (the world). In addition to taking considerably more time, training with physical agents is much more expensive, since you need to purchase the robot and account for any maintenance. It would be ideal if we could train such agents without having to give them physical bodies.

6.4.3. Simulators

Simulators address the preceding concerns. Instead of using an expensive robot or building a car with the necessary sensors, we could instead use computer software to emulate the experiences the environment would provide. For example, when training agents to drive autonomous cars, instead of equipping cars with the necessary sensors and deploying the model on physical cars, we could just train the agents inside software environments, such as the driving game Grand Theft Auto. The agent would receive as input the images of its surroundings, and it would be trained to output driving actions that would get the vehicle to the programmed destination as safely as possible.

Not only are simulators significantly cheaper to train agents with, but agents are able to train much more quickly since they can interact with the simulated environment much faster than in real life. If you need to watch and understand a two-hour movie, it will require two hours of your time. If you focus more intensely, you could probably increase the playback speed by two or three, dropping the amount of time needed to an hour or a bit less. A computer, on the other hand, could be finished before you've viewed the first act. For example, an 8 GPU computer (which could be rented from a cloud service) running ResNet-50, an established deep learning model for image classification, can process over 700 images per second. In a two-hour movie running at 24 frames per second (standard in Hollywood), there are 172,800 frames that need to be processed. This would require four minutes to finish. We could also effectively increase the playback speed for our deep learning model by dropping every few frames, which will drop our processing time to under two minutes. We could also throw more computers at the problem to increase processing power. For a more recent reinforcement learning example, the OpenAI Five bots were able to play 180 years of Dota 2 games